Why isn’t water a gas at normal temperatures?

Chapter 4 Intermolecular Forces

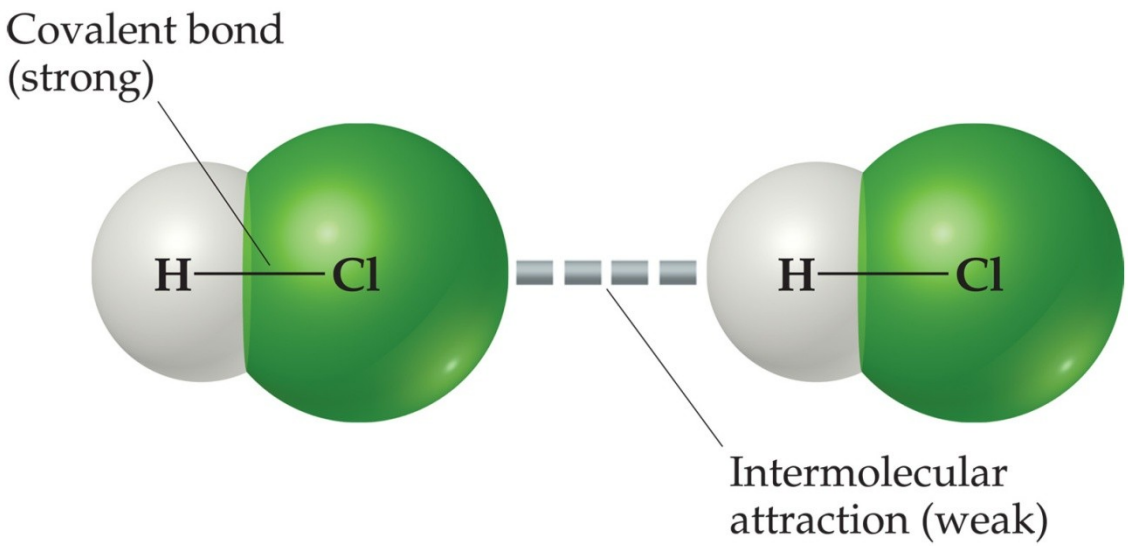


Figure 10.2

There are weak attractive forces between most molecules. These forces are typically 1-2 orders of magnitude weaker than the intramolecular attractions (e.g. the covalent bonds between atoms)

Intermolecular Forces

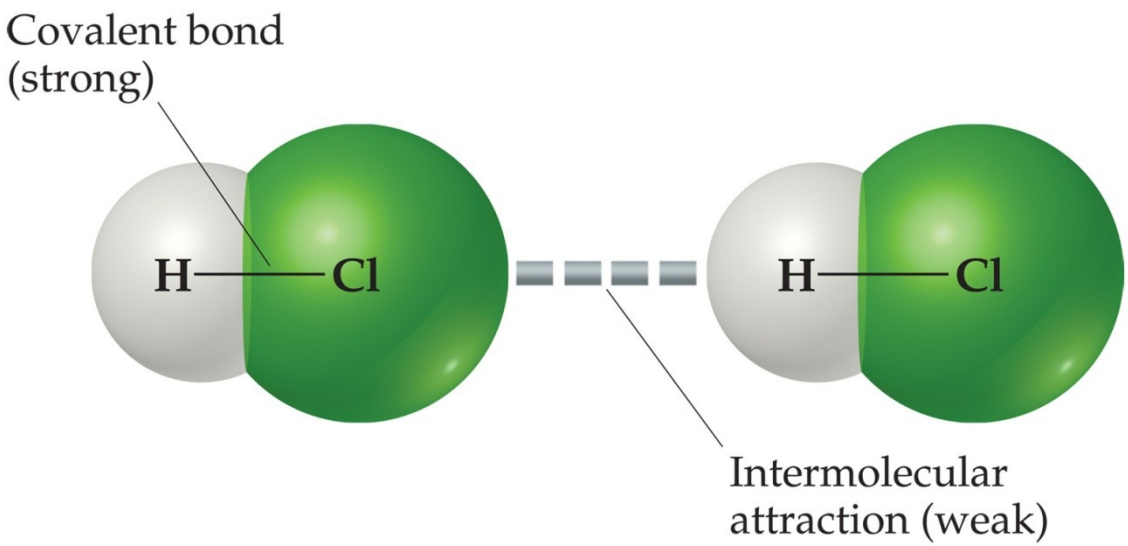
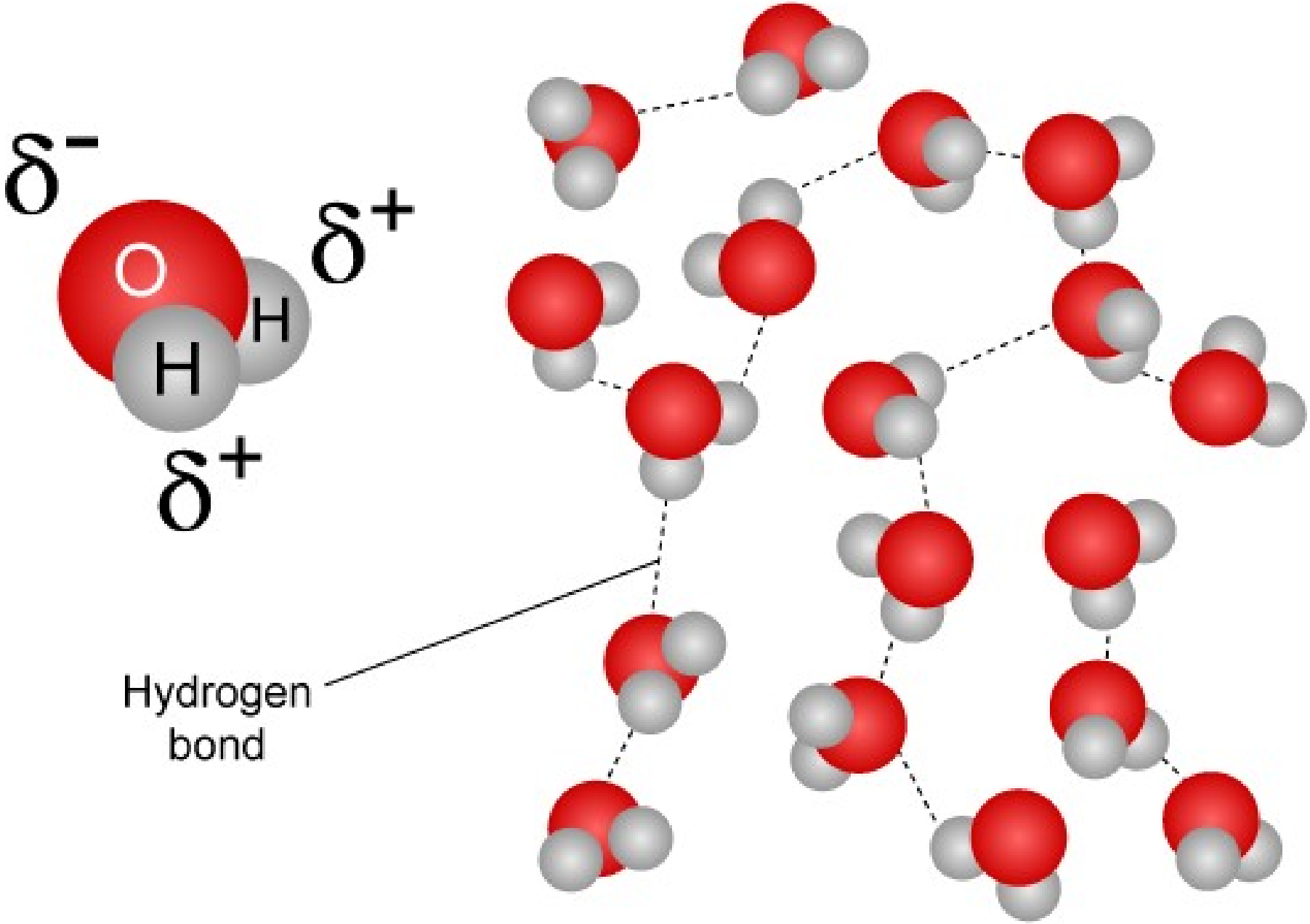


Figure 10.2

They are, however, strong enough to control physical properties such as boiling and melting points, vapour pressures, and viscosities.

Give it a thought!
What happens when water boils?
Which bonds break?

$H_2O(l) + \text{energy} \rightarrow H_2O\ (g)$



(length appears different for perspective (3D))

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https://sphweb.bumc.bu.edu/otlt/MPH-Modules/PH/PH709_BasicCellBiology/PH709_BasicCellBIology4.html

Intermolecular Forces

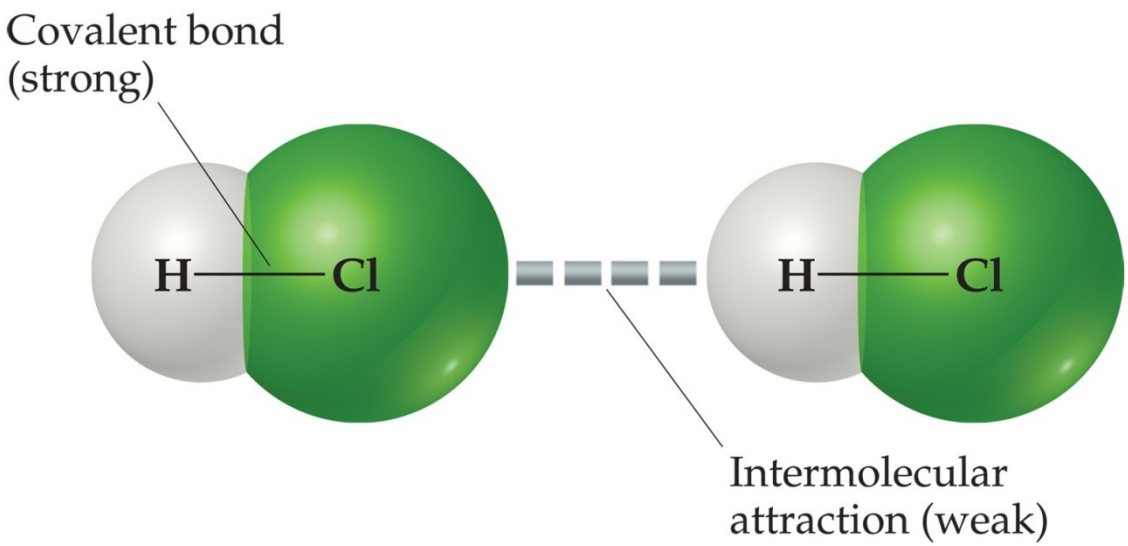
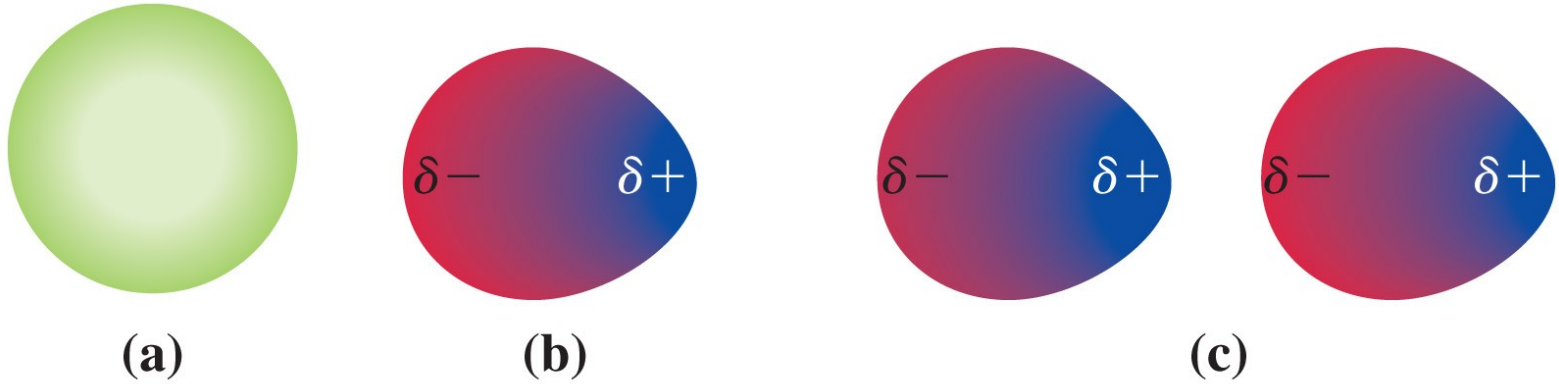


Figure 10.2

These intermolecular forces are referred to as van der Waals forces.

Intermolecular Forces

- Polar species have a permanent dipole moment (μ). Electrons in a nonpolar species may be displaced momentarily resulting in an **instantaneous dipole**.
- This can result in an **induced dipole** in nearby molecules, causing an attraction between the molecules called a **dispersion** or **London force**.



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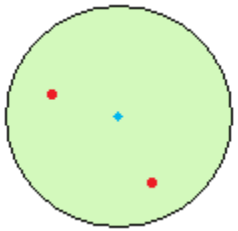
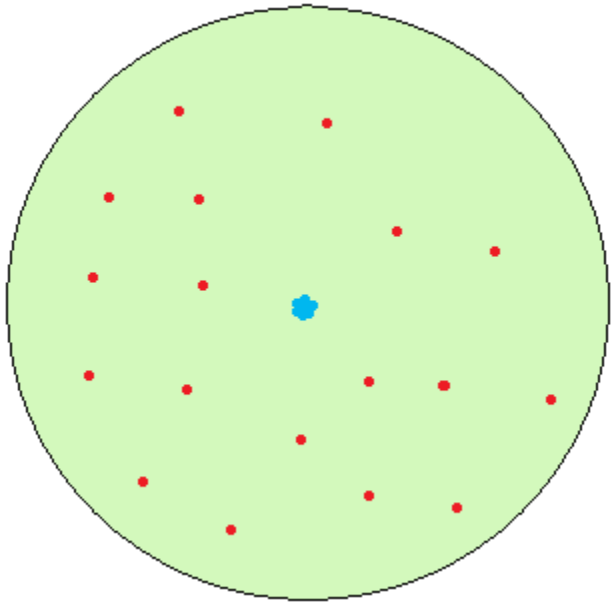
Dispersion forces

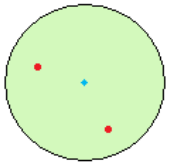
- Where do you see a greater likelihood for an spontaneous dipole in a helium or an argon atom?
- Which is more ***polarizable***

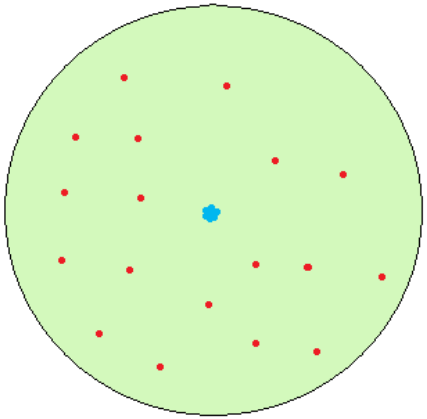




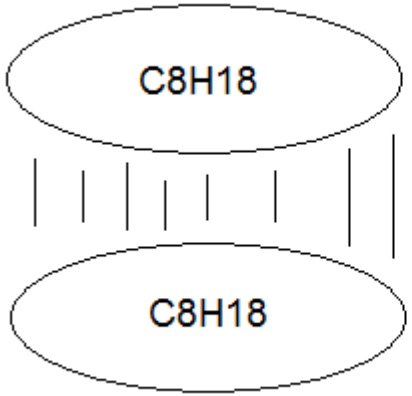
Dispersion forces

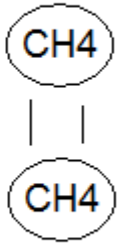
- We say that argon is more ***polarizable*** then helium.
- Consequences: melting and boiling points
- mp_{He} <1K mp_{Ar} 83.8 K
- bp_{He} 4.2 K bp_{Ar} 87.3 K





- The larger the molecules the more electrons, the greater the attraction. Careful it is not gravity!





- Octane b.p. 126 °C

Methane b.p. - 162 ° C

- More interactions exist between larger molecules

Intermolecular Force vs. Gravity

- For water the heat of vaporization is 40.8 kJ/ mol. How does this compare with the work that would need to be done to lift one mol of water by one metre?
- Work of lifting =

Intermolecular Force vs. Gravity

Work of lifting = m x g x h

= (0.018 kg) (9.81 m/s2)(1 m)

= 0.176 J

Compare that to 40800 J needed to evaporate one mole of water! It is not gravity keeping water in the ocean!

Molecular Shape

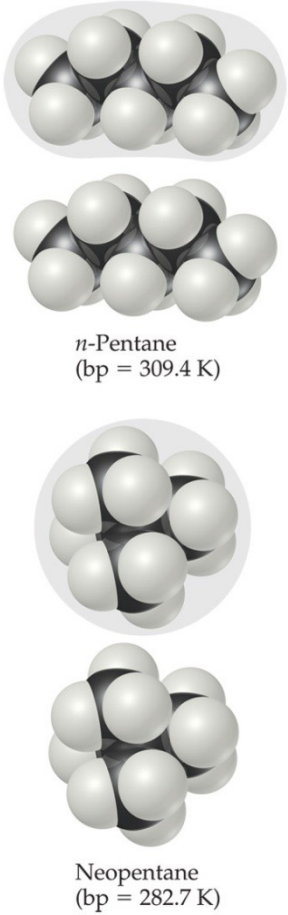
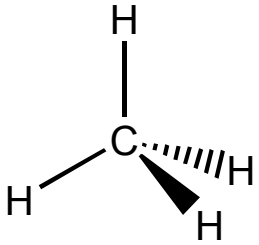


Figure 10.6

- The shape of the molecule affects the strength of dispersion forces: long, skinny molecules (like *n*-pentane tend to have stronger dispersion forces than short, fat ones (like neopentane).
- This is due to the increased surface area in *n*-pentane. It is also easier to polarize a long cigar shaped molecule than a sphere.

Important Summary

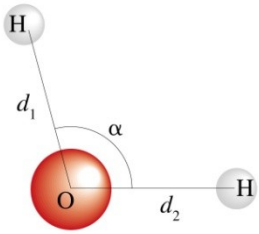
- When comparing molecules of comparable molecular mass usually the more polar molecule has the higher boiling and melting point.



methane CH_4 water H_2O

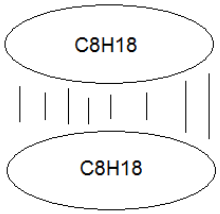
molecular mass 16 u molecular mass 18 u

Boiling point -162 °C Boiling point 100 °C



- When comparing molecules of vastly different molecular mass the one with the greater mass will have the higher melting and boiling point.

Octane b.p. 126 °C



Methane b.p. - 162 ° C

Example

Arrange in order of decreasing boiling point:

- He, Ne, Cl_2, $(CH_3)_2CO$, O_2

Example

Arrange in order of decreasing boiling point:

- He, Ne, Cl_2, $(CH_3)_2CO$, O_2

species	He	Ne	Cl_2	$(CH_3)_2CO$	O_2
Molar mass	4	20	71	58	32
polar	No	No	No	Yes	No

Example

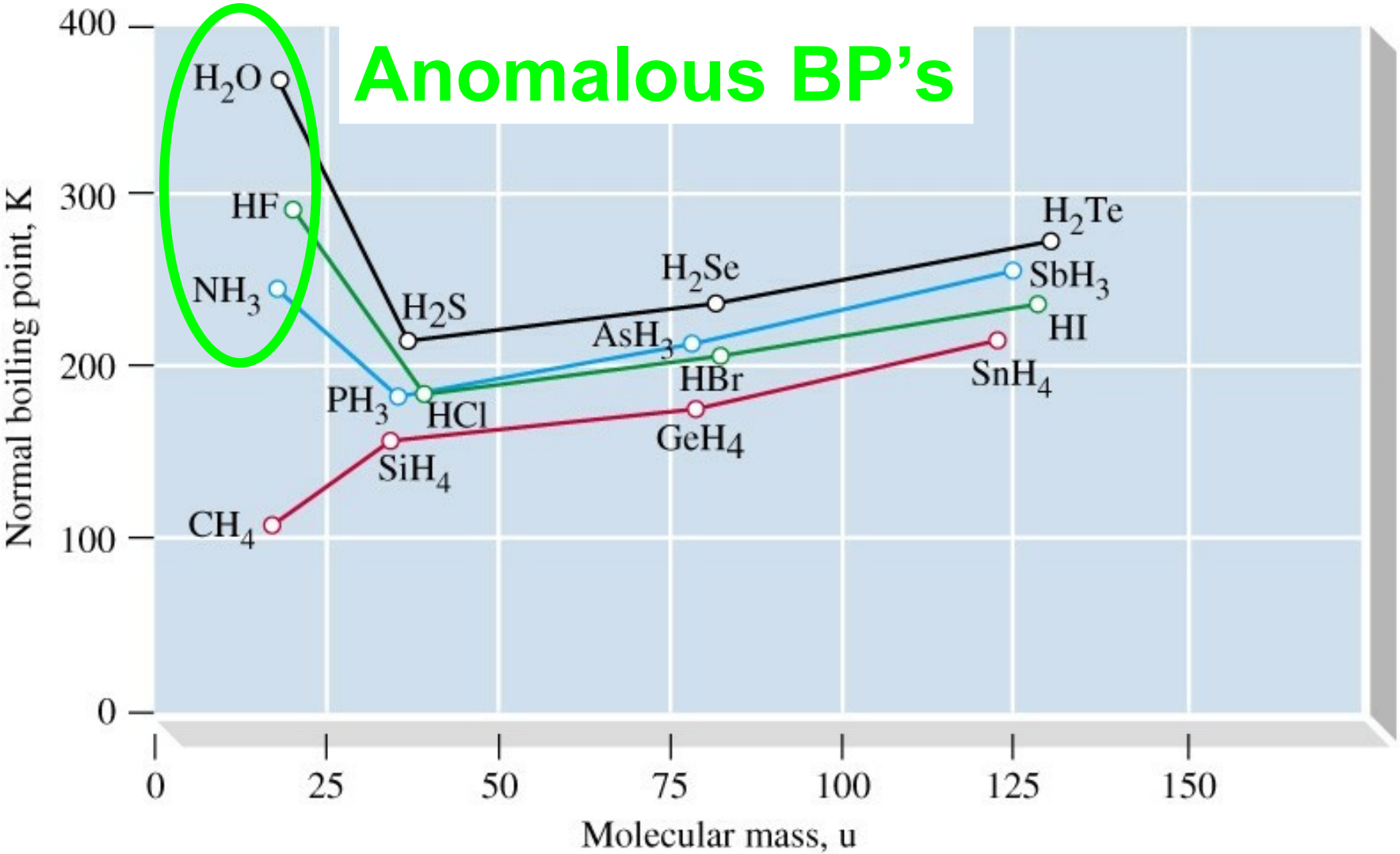
Arrange in order of decreasing boiling point:

- He, Ne, Cl_2, $(CH_3)_2CO$, O_2

<u>Answer</u>:

$(CH_3)_2CO_{(56)} > Cl_{2(-34)} > O_{2(-183)} > Ne_{(-246)} > He_{(-269)}$

Hydrogen Bonding



Hydrogen Bonding

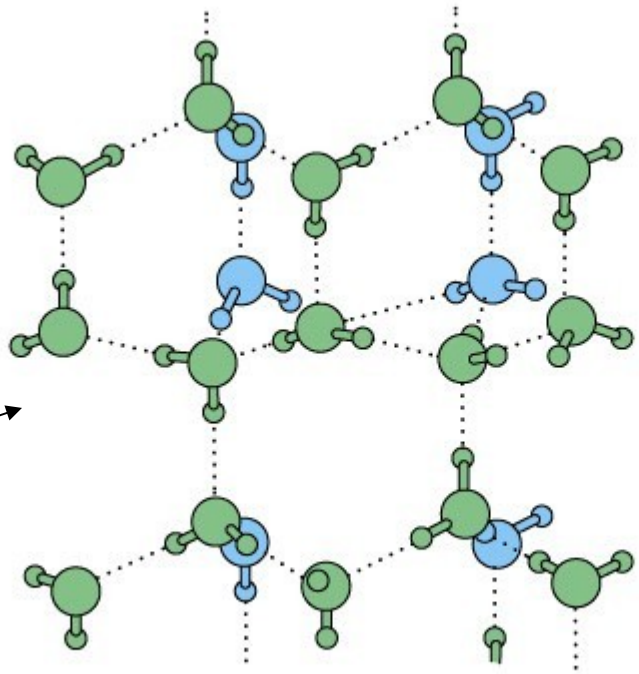
- The anomolous boiling points of NH_3, H_2O and HF are a result of **hydrogen bonding**, a special type of intermolecular force.

- Hydrogen bonding results when a hydrogen atom on a highly electronegative atom is attracted to a highly electronegative atom (usually F, O and N) on an adjacent molecule.

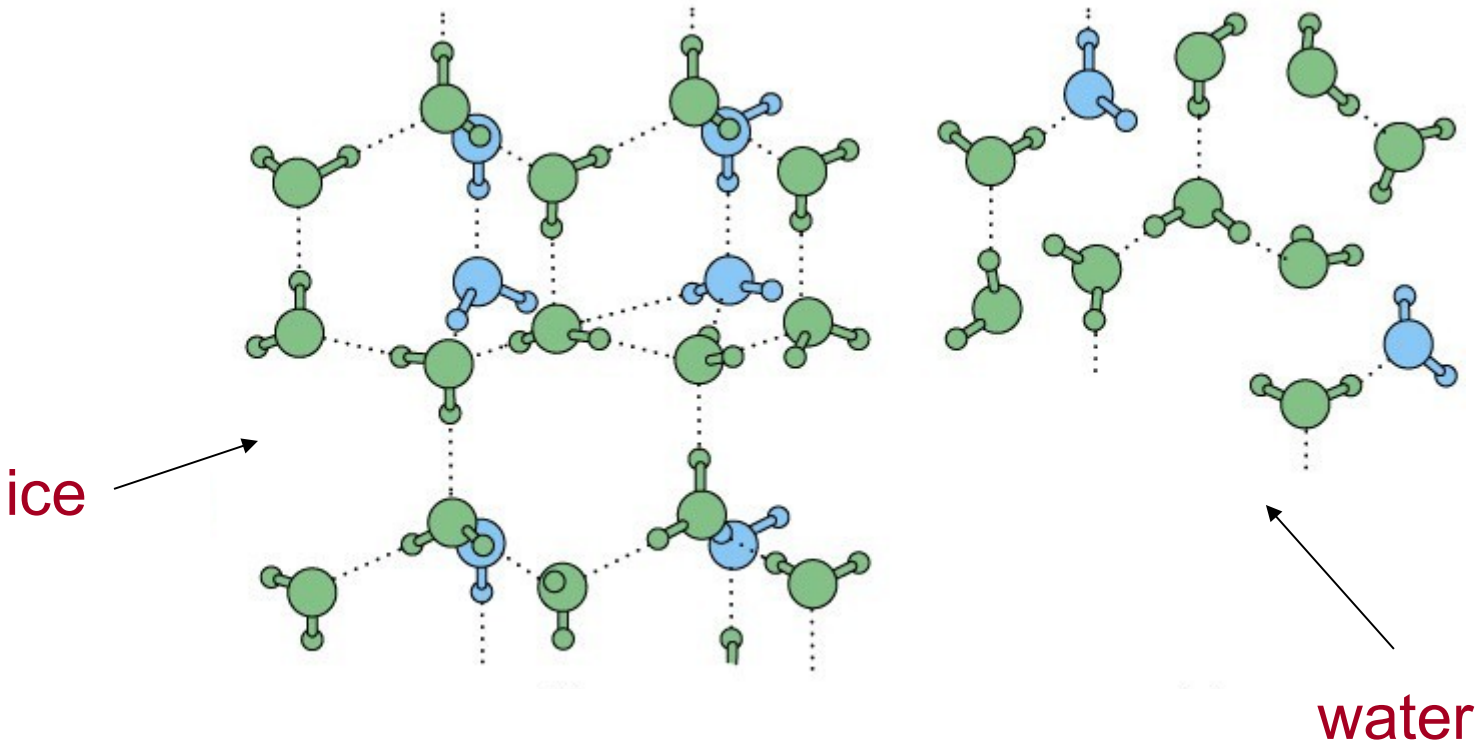
- Hydrogen bond energies are ~ 15 to 40 kJ mol^{-1}

Hydrogen Bonding in Water

ice



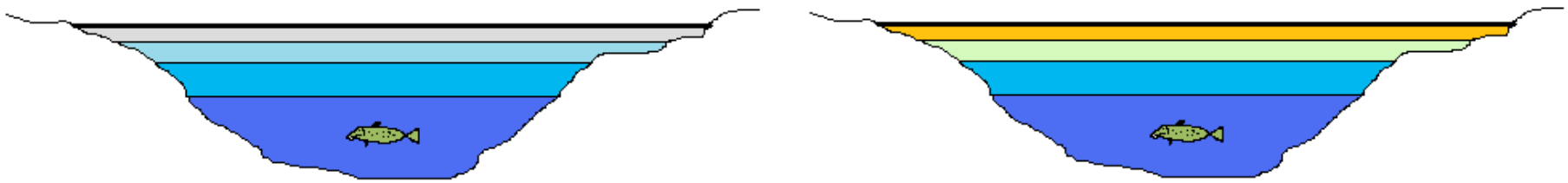
Hydrogen Bonding in Water



- Ice is rigid - less dense than liquid water; water can pack more densely because there are less hydrogen bonds in the liquid.

Consequences of anomalous properties

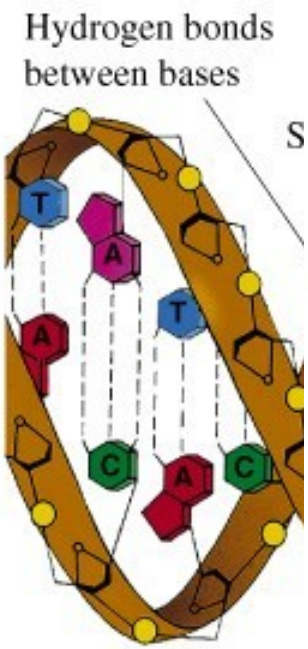
- Ice floats, water has its greatest density at 4°C



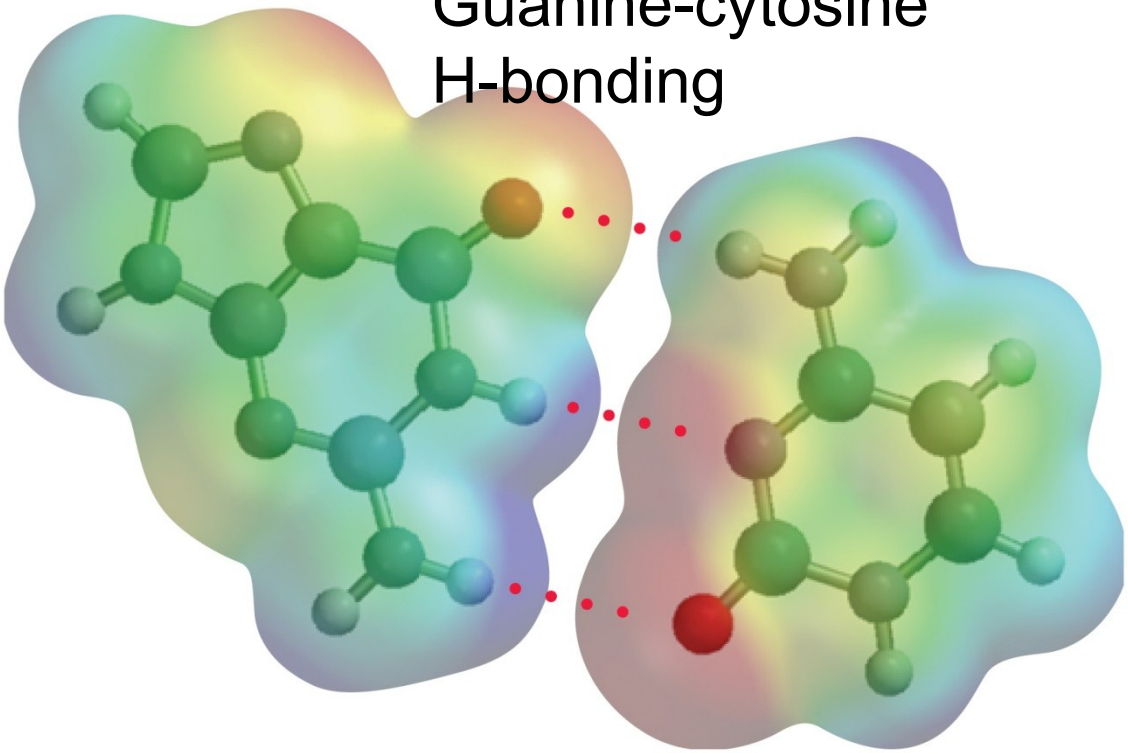
Winter or summer the bottom of the lake has approximately the same temperature of 4°C providing a stable environment.

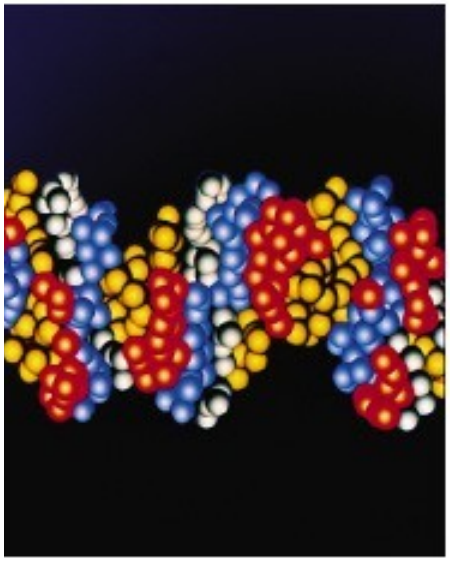
Hydrogen Bonding - an aside

so guess what holds together the double strands of DNA....



Guanine-cytosine
H-bonding





(b)

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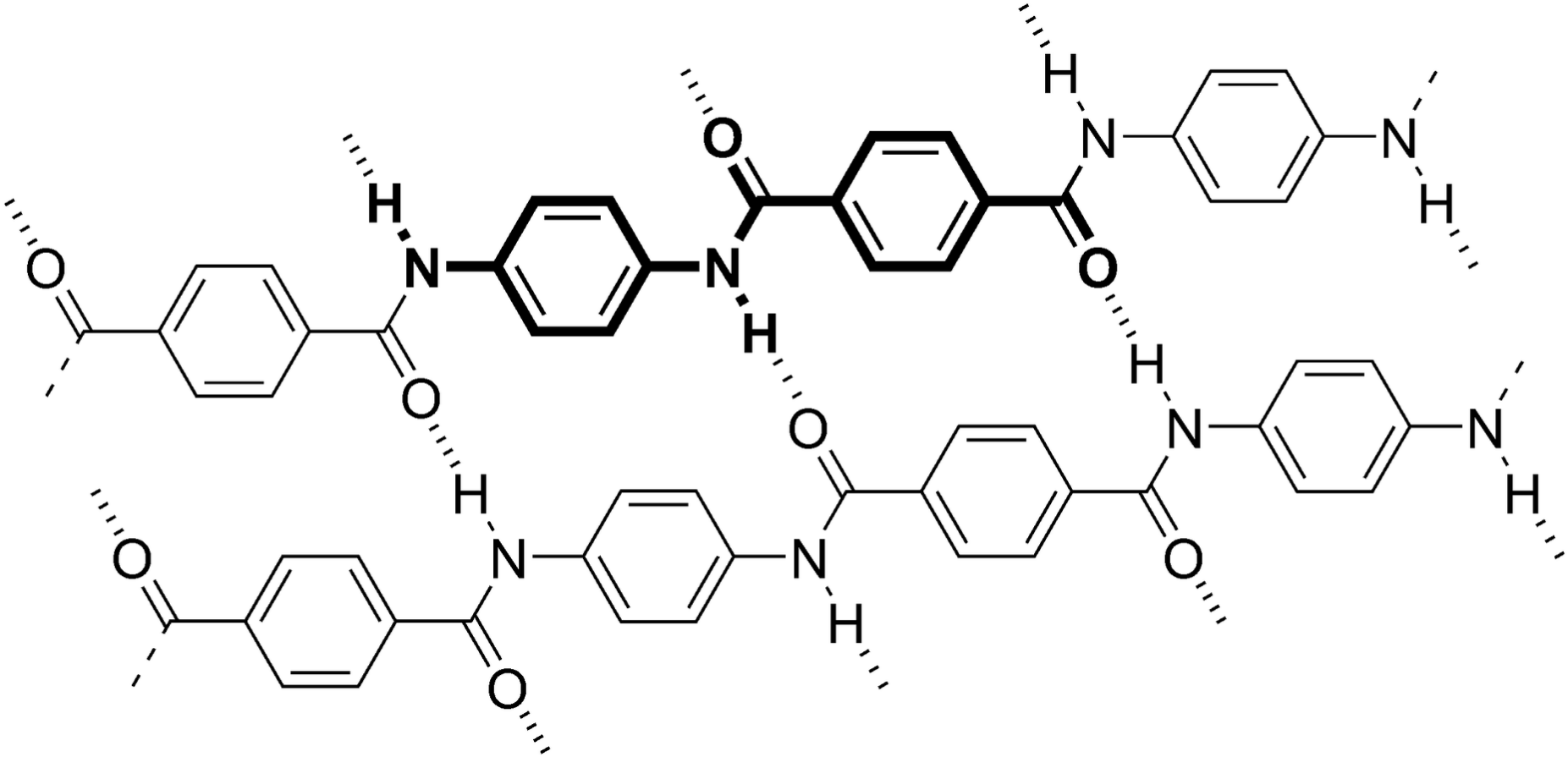
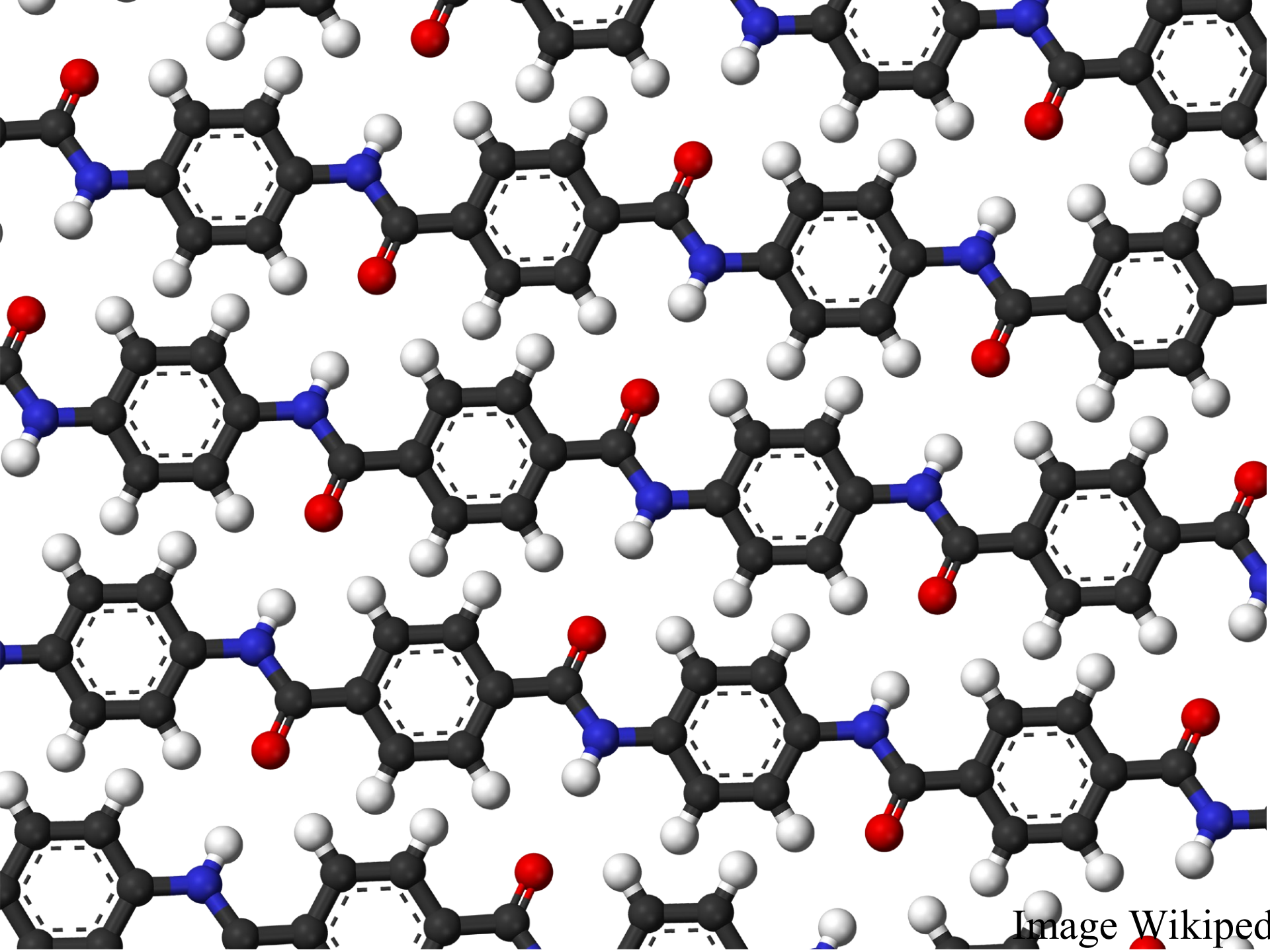
Hydrogen bonding in Kevlar, Image Wikipedia

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